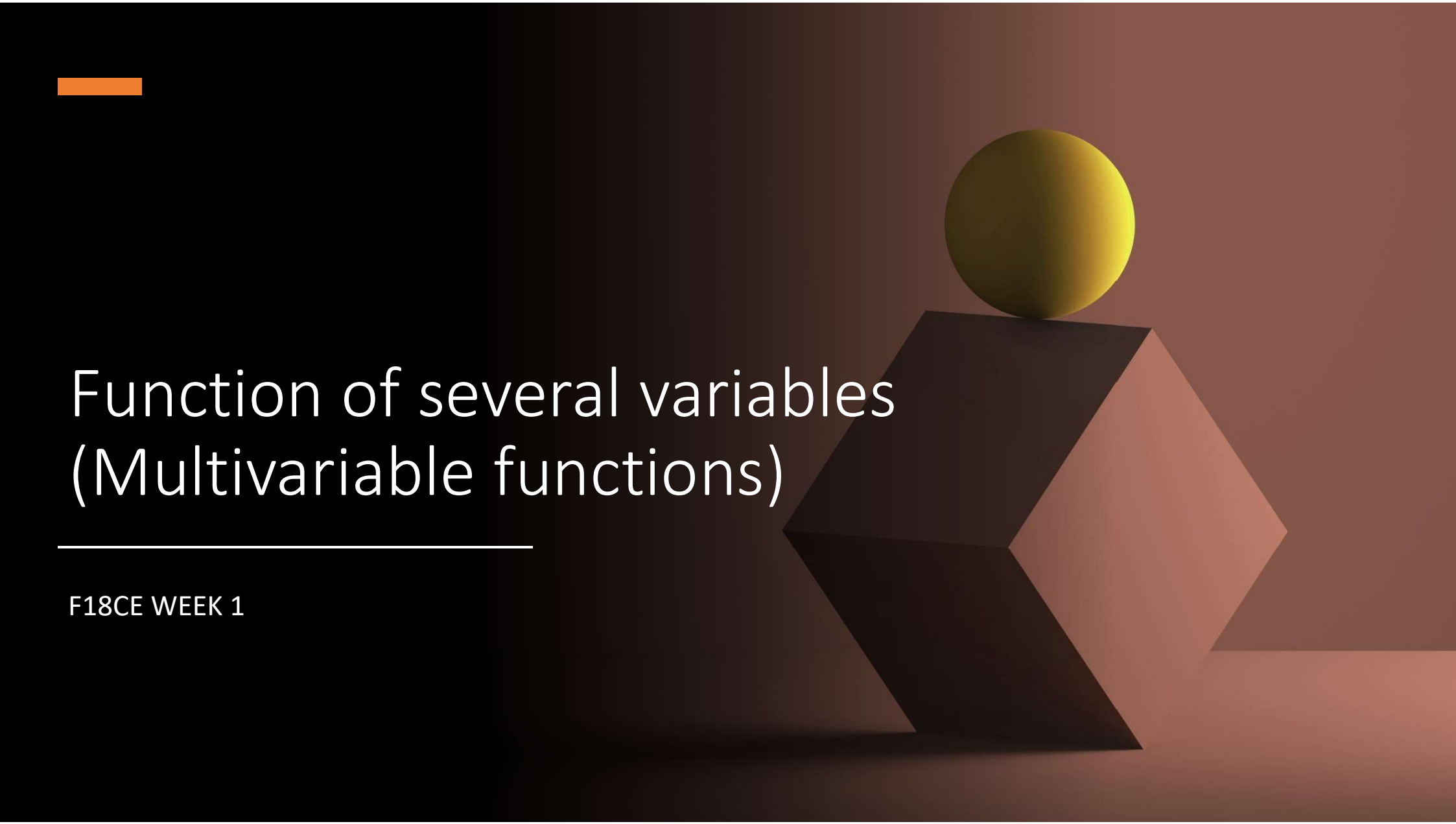
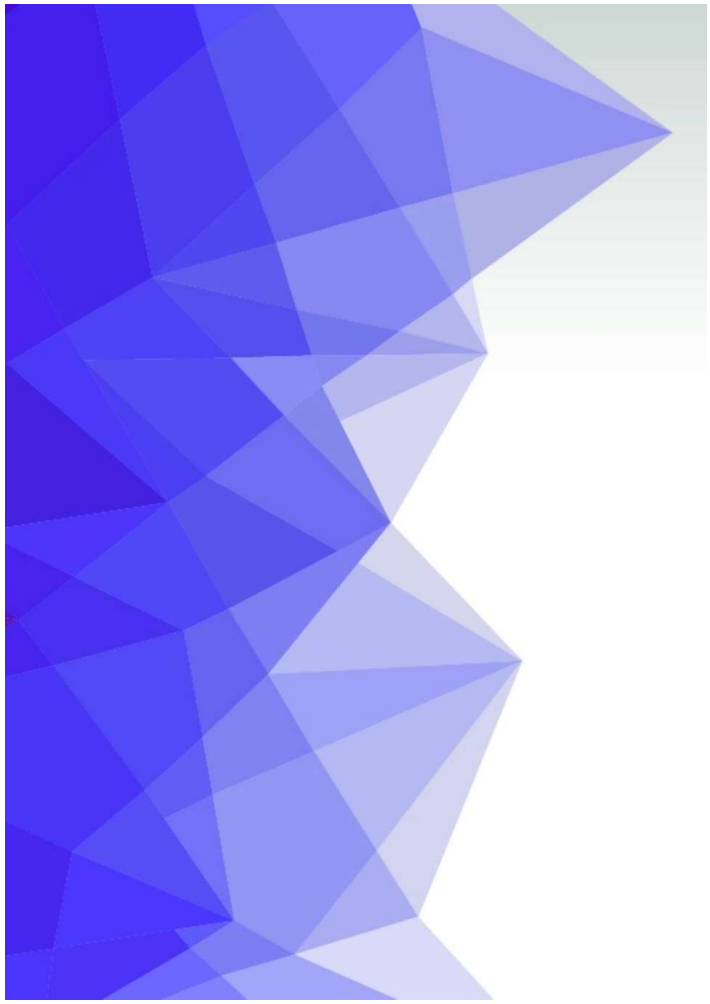


Function of several variables (Multivariable functions)

F18CE WEEK 1



Differentiation in multivariable function

Differentiation in several variables

Recall

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \text{ OR } \lim_{\Delta x \to 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Definition 2.1. Partial derivatives

Consider $f:\mathbb{R}^2\to\mathbb{R}$ and let $\left(a,b\right)\in\mathbb{R}^2$. The partial derivative of f at $\left(a,b\right)$, with respect to x, is denoted as $\dfrac{\partial f}{\partial x}\left(a,b\right)$, or $f_x\left(a,b\right)$ is defined as

$$\frac{\partial f}{\partial x}=\lim_{\Delta x\to 0}\frac{f(x+\Delta x,y)-f(x,y)}{\Delta x}.$$

When $\left(x,y\right)=\left(a,b\right)$ substituted by a specific point, then we have

$\dfrac{\partial f}{\partial x}\left(a,b\right)=\lim\limits_{\Delta x\to 0}\dfrac{f(a+\Delta x,b)-f(a,b)}{\Delta x}$. Remark: We can also write $\left.\dfrac{\partial f}{\partial x}\right|_{(a,b)}$ as $\dfrac{\partial f}{\partial x}(a,b)$.

Similarly, the partial derivative of f with respect to y, is denoted as $\dfrac{\partial f}{\partial y}$ or f_y, is defined as

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}.$$

Definition for partial derivatives of a multivariable function

Definition. Suppose $f(x_1, x_2, \ldots, x_n) : \mathbb{R}^n \to \mathbb{R}$ is a function.

Then $$\frac{\partial f}{\partial x_i} = \lim_{h \to 0} \frac{f(x_1, \ldots, x_{i-1}, x_i + h, x_{i+1}, \ldots, x_n) - f(x_1, \ldots, x_n)}{h} \text{ for all } i = 1, \ldots, n.$$

Thus, $$\frac{\partial f}{\partial x_i}(a_1, \ldots, a_n) = \lim_{h \to 0} \frac{f(a_1, \ldots, a_{i-1}, a_i + h, a_{i+1}, \ldots, a_n) - f(a_1, \ldots, a_n)}{h}.$$

Example 2.2

Let $f(x,y) = x^2 + y^3$. Find f_x and f_y.

$$\frac{\partial f}{\partial x} = \lim_{h\to 0} \frac{f(x+h,y)-f(x,y)}{h}$$
$$= \lim_{h\to 0} \frac{(x+h)^2+y^3-(x^2+y^3)}{h}$$
$$= \lim_{h\to 0} \frac{2hx+h^2}{h}$$
$$= \lim_{h\to 0}(2x+h) = 2x$$

$$\frac{\partial f}{\partial y} = \lim_{h\to 0} \frac{f(x,y+h)-f(x,y)}{h}$$
$$= \lim_{h\to 0} \frac{x^2+(y+h)^3-(x^2+y^3)}{h}$$
$$= \lim_{h\to 0} \frac{3y^2h+3yh^2+h^3}{h}$$
$$= \lim_{h\to 0}(3y^2+3yh+h^2) = 3y^2$$

We normally will not be using the definition to find the partial derivative if it is trivial.

Example 2.3

Let $f(x,y) = x^3 + 3xy + y^2$ and $g(x,y) = \frac{x}{y}$.

Find f_x and f_y; and g_x and g_y.

Example 2.4

Let $f(x_1, x_2, x_3, x_4) = x_1^2 + x_2^3 + x_3^4 + x_4^5$ and $g(x, y, z) = xy \ln z$.

Find f_{x_3} and g_z.

Example 2.5

$f(x,y) = x^y,\ \text{where}\ \ x,y > 0\,.\ \text{Find}\ \ f_x\ \ \text{and}\ \ f_y\,.$

Example 2.6

Let $f(x,y) = \begin{cases} \frac{2x^3+xy+4y^4}{\sin^2(2x)+\sin^2(3y)} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$.

Evaluate $\left.\frac{\partial f}{\partial x}\right|_{(0,0)}$ and $\left.\frac{\partial f}{\partial y}\right|_{(0,0)}$ if they exist.

$$
\begin{aligned}
\left.\frac{\partial f}{\partial x}\right|_{(0,0)} &= \lim_{h\to 0} \frac{f(0+h,0)-f(0,0)}{h} \\
&= \lim_{h\to 0} \frac{f(h,0)-0}{h} \\
&= \lim_{h\to 0} \frac{2h^3}{h\sin^2(2h)} \\
&= \frac{1}{2}\lim_{h\to 0}\left(\frac{2h}{\sin(2h)}\right)^2 \\
&= \frac{1}{2}
\end{aligned}
$$

$$
\begin{aligned}
\left.\frac{\partial f}{\partial y}\right|_{(0,0)} &= \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} \\
&= \lim_{h\to 0} \frac{4h^4}{h\sin^2(3h)} \\
&= \frac{4}{9}\lim_{h\to 0} \frac{9h^3}{\sin^2(3h)} \\
&= \frac{4}{9}\lim_{h\to 0} \frac{9h^2}{\sin^2(3h)} \times \lim_{h\to 0} h \\
&= \frac{4}{9}\times 1 \times 0 = 0
\end{aligned}
$$

Remark

Theorem in F18CD:

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable at c.

Then f is continuous at c.

Question in F18CE:

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a partially differentiable at $(a,\ b)$.

Does it follow that f is continuous at $(a,\ b)$?

A counter example

Let $f(x,y) = \begin{cases} \frac{x^2}{x-y}, & x \neq y \\ 0\ , & x = y \end{cases}$. Note that f is not continuous at $(0,\ 0)$

(take pathways $y = x \pm x^2$, or simply just $y = x - x^3$)

but partial derivatives $f_x(0,\ 0)$ and $f_y(0,\ 0)$ perfectly well defined.

$$f_x(0,0) = \lim_{h\to 0} \frac{f(0+h,0)-f(0,0)}{h} = \lim_{h\to 0} \frac{\left(\frac{h^2}{h-0}\right)-0}{h} = 1$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,0+h)-f(0,0)}{h} = \lim_{h\to 0} \frac{\left(\frac{0^2}{0-h}\right)-0}{h} = 0.$$

Extra example (a)

The equation $z^5 + y^2e^z + e^{2x} = 0$ implicitly determines z as a function of x and y, i.e., $z(x, y)$ (x and y are independent and z is dependent). For example, when $x = y = 0$, the equation reduces to $(z(0,0))^5 = -1$ which implies that $z(0,0) = -1$ as $(z(0,0))^4 \geq 0$. Evaluate $\frac{\partial z}{\partial x}(0,0)$.

Solution. Implicitly & partially differentiate $z^5 + y^2e^z + e^{2x} = 0$ with respect to x : $5z^4 \frac{\partial z}{\partial x} + y^2e^z \frac{\partial z}{\partial x} + 2e^{2x} = 0 \Rightarrow \frac{\partial z}{\partial x} = -\frac{2e^{2x}}{5z^4+y^2e^z}$.

Hence, $\frac{\partial z}{\partial x}(0,0) = -\frac{2}{5(-1)^4+0} = -\frac{2}{5}$.

Extra example (b)

Evaluate $\lim_{z\to 0} \frac{(x+y+z)^3-(x+y)^3}{(x+y)z}$.

Solution. Let $f(u,v,w) = \frac{(u+v+w)^3}{u+v}$. Then (treat z as h)

$$\begin{aligned}
\lim_{z\to 0} \frac{(x+y+z)^3-(x+y)^3}{(x+y)z} &= \lim_{z\to 0} \frac{f(x,y,0+z)-f(x,y,0)}{z} \\
&= \frac{\partial f}{\partial w}(x,y,0) \\
&= \left.\frac{3(u+v+w)^2(1)-0}{u+v}\right|_{(x,y,0)} \\
&= 3(x+y)
\end{aligned}$$

You can even simplify this to 2-variable function (see that x + y)

A theorem in single variable Calculus

Theorem. Let $y(x) : \mathbb{R} \to \mathbb{R}$ be an invertible function

(so that $x(y) : \mathbb{R} \to \mathbb{R}$ is defined)

such that $\frac{dy}{dx}$ and $\frac{dx}{dy} \neq 0$ exist, then $\frac{dy}{dx} = \left(\frac{dx}{dy}\right)^{-1}$.

This result is not true in general when we have partial derivative.

A counter example

Let $x(r,\theta) = r\cos\theta, y(r,\theta) = r\sin\theta$.

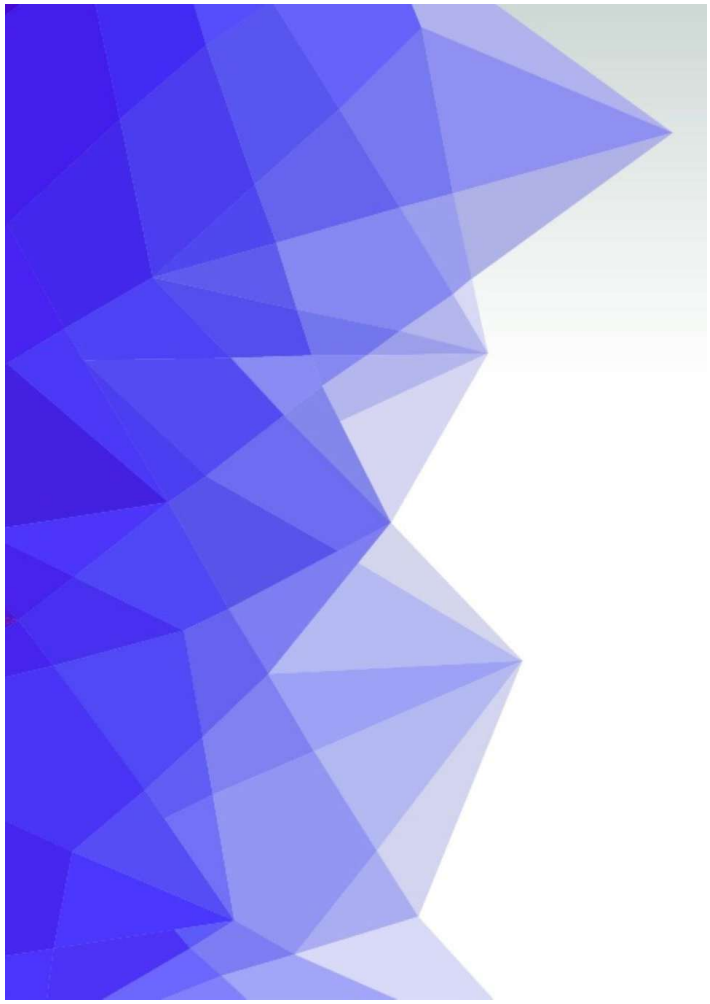
Then $r(x,y) = \sqrt{x^2+y^2}$.

We show that $\frac{\partial x}{\partial r} \neq 1 \div \frac{\partial r}{\partial x}$ in general.

Proof. $\frac{\partial x}{\partial r} = \cos\theta$ and $\frac{\partial r}{\partial x} = \frac{2x}{2\sqrt{x^2+y^2}} = \frac{r\cos\theta}{r} = \cos\theta$.

It is clear that $\cos\theta \neq \frac{1}{\cos\theta}$ for all θ .

In fact, $\cos\theta = \frac{1}{\cos\theta} \Leftrightarrow \cos^2\theta = 1 \Leftrightarrow \theta = n\pi, n \in \mathbb{Z}$.



Differentiation in multivariable – Rigorous definition

Differentiability of multivariable functions

1. Recall $y = mx + c$ where m is the gradient and c is the y-intercept.
2. A more general form: $y - y_0 = m(x - x_0)$, where (x_0, y_0) is a point on the line.
3. m is the gradient and can also be seen as $\dfrac{\Delta y}{\Delta x}$, or $\dfrac{f(a) - f(b)}{a - b}$, or $\lim\limits_{a \to b} \dfrac{f(a) - f(b)}{a - b}$,

this, as you know, leads to the definition of $\lim\limits_{x \to a} \dfrac{f(x) - f(a)}{x - a}$.

4. Rewrite with $y = f(x)$, $y_0 = f(x_0)$ and $m = f'(x_0)$, we then have

$$\boxed{f(x) = f(x_0) + f'(x_0)(x - x_0)}$$

5. When f is a constant or linear, the above formula is accurately equal for all x.

6. When f is of higher degree, it is called a *linear approximation* for f at $x = x_0$.

7. Since, it is an approximation, we replace $f(x)$ as $L_a(x)$, $a = x_0$ and write

$$\boxed{L_a(x) = f(a) + f'(a)(x - a)}$$

8. Obviously $L_a(x) \approx f(x)$ is an approximation.

9. A good approximation means that the error $\left|f(x) - L_a(x)\right|$ is small. The error is controlled by how close x to a (the closer the lesser the error).

10. Suppose f is differentiable at $x = a$ (i.e., $f'(a) \exists$). Then

$$
\begin{aligned}
0 &= f'(a) - f'(a) \\
&= \lim_{x \to a} \left[\frac{f(x) - f(a)}{x - a} - f'(a) \right] \\
&= \lim_{x \to a} \frac{f(x) - f(a) - f'(a)(x - a)}{x - a} \\
&= \lim_{x \to a} \frac{f(x) - L_a(x)}{x - a}
\end{aligned}
$$

11. This motivates us to define:

Definition 2.13

f is differentiable at $x = a$ if (a) $f'(a) \exists$ and (b) $\lim_{x \to a} \dfrac{f(x) - L_a(x)}{x - a} = 0$.

12. The Definition 2.13 helps us to define the differentiability of function with more than one variable. Now based on Definition 2.1, suppose $f_x(a,b)$ and $f_y(a,b)$ exist. Then we can define

$$L_{\underset{\sim}{v}}(x,y) = f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b)\,,$$

where $\underset{\sim}{v} = (a,b)\,.$

13. We can now define the differentiability of a function with more than one variable at point $\underset{\sim}{v} = (a,b)\,.$

Definitions 2.14, 2.15

Definition 2.14. $f(x,y)$ is differentiable at $\underset{\sim}{v} = (a,b)$ if

$$\text{(a) } f_x(a,b) \text{ and } f_y(a,b) \ \exists \text{ and (b) } \lim_{(x,y)\to(a,b)} \frac{f(x,y) - L_{\underset{\sim}{v}}(x,y)}{\sqrt{(x-a)^2 + (y-b)^2}} = 0$$

Definition 2.15. Let $D_f \subseteq \mathbb{R}^2$. Then $f : D_f \to \mathbb{R}$ $f(x,y)$ is said to differentiable if $f(x,y)$ is differentiable at every point $\underset{\sim}{v} = (a,b) \in D_f$.

Note: $L_{\underset{\sim}{v}}(x,y) = f(a,b) + (x-a)f_x(a,b) + (y-b)f_y(a,b)$

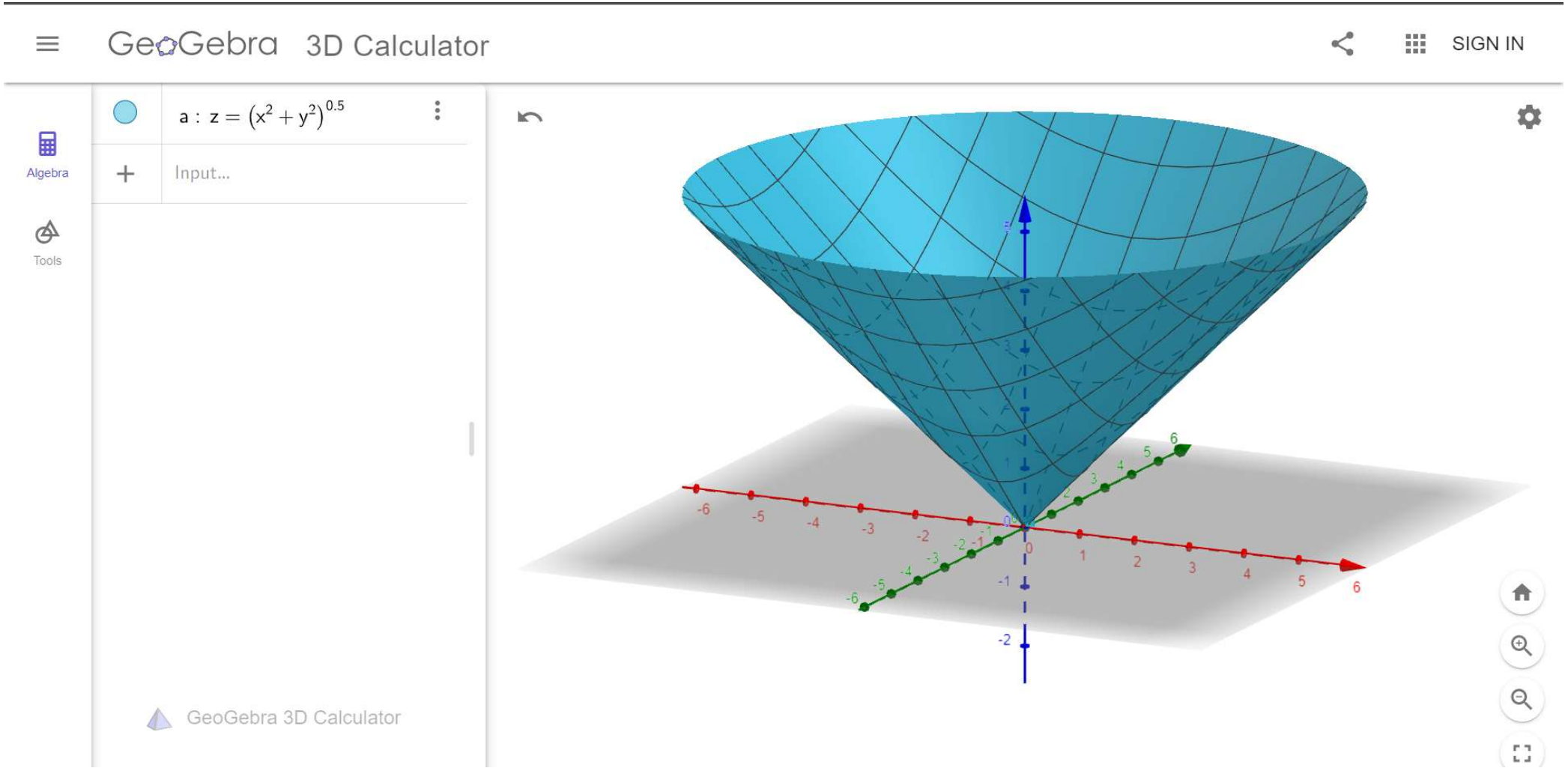
Example 2.16

Example 2.16 Show that $f(x,y)=\sqrt{x^2+y^2}$ is not differentiable at (0, 0).

Proof. We apply the definition to show that $f_x(a,b)$ does not exist at (0, 0).

$$\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim_{h\to 0}\frac{\sqrt{h^2}-0}{h}=\lim_{h\to 0}\frac{|h|-0}{h}\nexists \because \lim_{h\to 0^-}\frac{-h}{h}=-1\neq 1=\lim_{h\to 0^+}\frac{h}{h}$$

Similarly, we can show that $f_y(a,b)=\lim_{h\to 0}\frac{f(a,b+h)-f(a,b)}{h}$ does not exist. Hence, $f(x,y)=\sqrt{x^2+y^2}$ is not differentiable at (0, 0).


GeoGebra 3D Calculator
SIGN IN
Algebra
Tools
a : z = (x² + y²)^0.5
Input...
GeoGebra 3D Calculator

Example 2.17

Show that $g(x,y) = -|x|$ and $f(x,y) = |x|$ are not differentiable at (0,0).

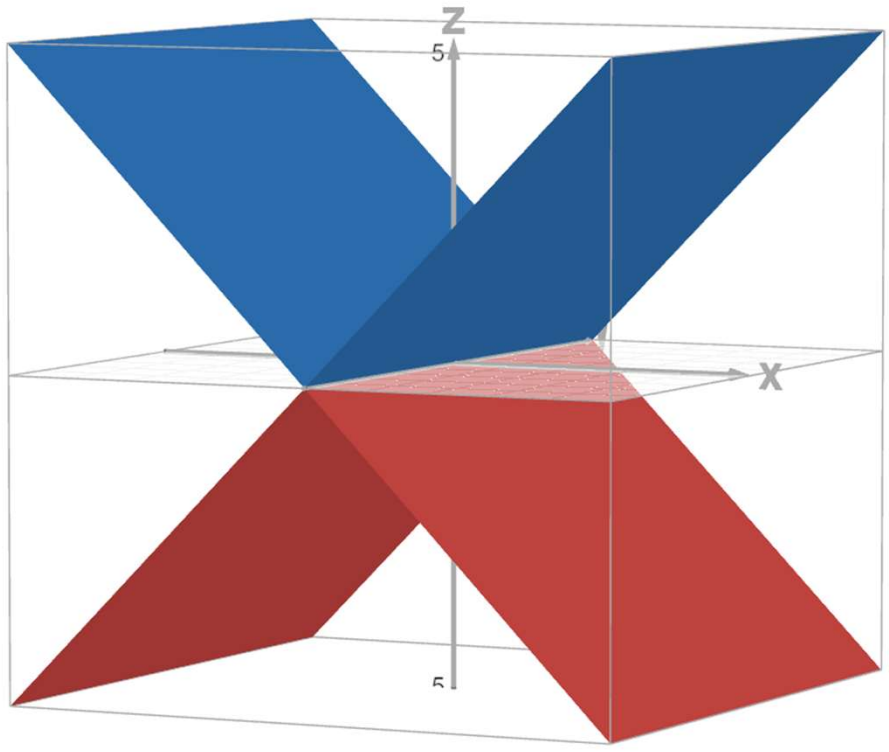
Proof. We show that $g_x(0,0)$ and $f_x(0,0)$ do not exist.

By definition, $\lim_{h\to\infty} \frac{g(h,0)-g(0,0)}{h} = \lim_{h\to\infty} \frac{-|h|-0}{h} \nexists \because \lim_{h\to 0^-} \frac{h}{h} = 1 \neq -1 = \lim_{h\to 0^+} \frac{-h}{h}$, and

$$\lim_{h\to\infty} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to\infty} \frac{|h|-0}{h} \nexists \because \lim_{h\to 0^-} \frac{-h}{h} = -1 \neq 1 = \lim_{h\to 0^+} \frac{h}{h}.$$

Hence, both functions are not differentiable at (0,0).

1. $z = -|x|$
2. $z = |x|$
3.



Example 2.18

Show that $p(x,y) = (xy)^{1/3}$ is not differentiable at (0,0).

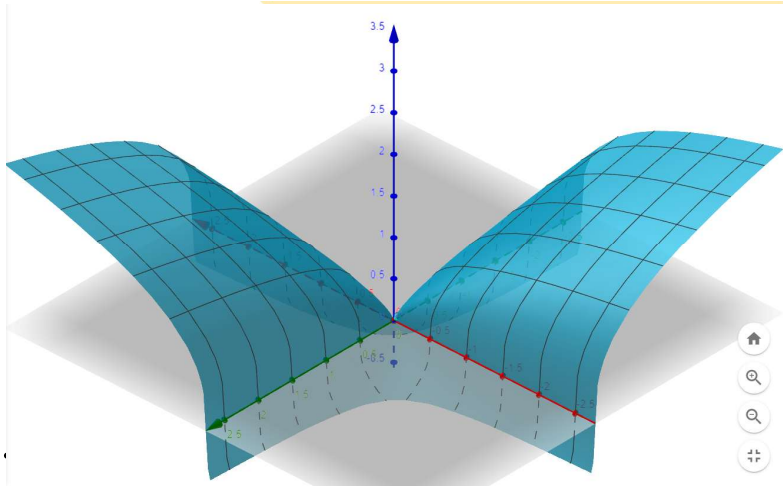
Proof. Check (a) $p_x(0,0) = \lim_{h\to 0} \frac{p(h,0)-p(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$

$$p_y(0,0) = \lim_{h\to 0} \frac{p(0,h)-p(0,0)}{h} = \lim_{h\to 0} \frac{0-0}{h} = 0$$

So $p_x = \begin{cases} \left(\frac{y}{3x^2}\right)^{1/3}, (x,y) \neq (0,0) \\ 0, \quad (x,y) = (0,0) \end{cases}$ and $p_y = \begin{cases} \left(\frac{x}{3y^2}\right)^{1/3}, (x,y) \neq (0,0) \\ 0, \quad (x,y) = (0,0) \end{cases}$ exist.

Now $L_{(0,0)}(x,y) = p(0,0) + (x-0)p_x(0,0) + (y-0)p_y(0,0) = 0$.

Check (b) $\lim_{(x,y)\to(0,0)} \frac{p(x,y)-L_{(0,0)}(x,y)}{\sqrt{(x-0)^2+(y-0)^2}} = \lim_{(x,y)\to(0,0)} \frac{(xy)^{1/3}-0}{\sqrt{x^2+y^2}}$. Let $G(x,y) = \frac{(xy)^{1/3}-0}{\sqrt{x^2+y^2}}$.



Now consider $\left(\frac{1}{n},\frac{1}{n}\right)\to\left(0,0\right)$, $\left(0,\frac{1}{n}\right)\to\left(0,0\right)$ as $n\to\infty$.

$\lim\limits_{(x,y)\to(0,0)} G(x,y)=\lim\limits_{n\to\infty} G\left(0,\frac{1}{n}\right)=\lim\limits_{n\to\infty} 0=0$ which is different from

$$\lim_{(x,y)\to(0,0)} G(x,y)=\lim_{n\to\infty} G\left(\frac{1}{n},\frac{1}{n}\right)=\lim_{n\to\infty}\frac{\left(\frac{1}{n}\right)^{2/3}}{\sqrt{\frac{2}{n^2}}}=\lim_{n\to\infty}\frac{n}{n^{2/3}\sqrt{2}}=\lim_{n\to\infty}\frac{n^{1/3}}{\sqrt{2}}=\infty\,.$$

Therefore, we conclude that $\lim\limits_{(x,y)\to(0,0)} G(x,y)$ does not exist.

Theorem 2.19, Corollary 2.20 & Remark

Theorem 2.19. Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function. If f is differentiable at (a,b), then it is continuous at (a,b).

Corollary 2.20. If f is not continuous at (a,b), then it is not differentiable at (a,b).

Remark. f may not be continuous at a point, even if all its partial derivatives exist at that point.

Example 2.21

Let $f:\mathbb{R}^2 \to \mathbb{R}$ be a function defined by $f(x,y)=\begin{cases} -\frac{3xy}{x^2+y^2}, (x,y)\neq(0,0) \\ 0, \quad (x,y)=(0,0) \end{cases}$.

Show that $f_x(0,0)$ and $f_y(0,0)$ exist but f is not differentiable at (0, 0).

Proof. By definition, $\begin{cases} f_x(0,0)=\lim\limits_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim\limits_{h\to 0}\frac{0-0}{h}=0 \\ f_y(0,0)=\lim\limits_{h\to 0}\frac{f(0,h)-f(0,0)}{h}=\lim\limits_{h\to 0}\frac{0-0}{h}=0 \end{cases}$ so both $f_x(0,0)$ and $f_y(0,0)$ exist. Consider pathways $y=\pm x$, we get

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{x\to 0} f(x,x)=-\tfrac{3}{2}\neq\tfrac{3}{2}=\lim_{x\to 0} f(x,-x)=\lim_{(x,y)\to(0,0)} f(x,y).$$

Thus, $\lim\limits_{(x,y)\to(0,0)} f(x,y)\not\exists \Rightarrow f$ is not continuous at (0, 0)$\Rightarrow f$ is not differentiable at (0, 0).

Theorem 2.22

Let $f : D \to \mathbb{R}$ be a function, where $D \subset \mathbb{R}^2$.

Let $(a, b) \in D$. Suppose

(a) f_x and f_y exist in some open disc containing (a, b); and

(b) f_x and f_y are continuous at (a, b).

Then f is differentiable at (a, b).

Final exam 2023/24

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function given by $f(x,y) = \frac{x^2 y}{x^4+2y^2}$.

Prove that f is differentiable at $\mathbb{R}^2 \setminus \{(0,0)\}$.

Proof. We see that both partial derivatives

$f_x = \frac{2xy(x^4+2y^2)-x^2y(4x^3)}{(x^4+2y^2)^2}$ and $f_y = \frac{x^2(x^4+2y^2)-x^2y(4y)}{(x^4+2y^2)^2}$

exist on $\mathbb{R}^2 \setminus \{(0,0)\}$. Since f_x and f_y are

continuous for all points on $\mathbb{R}^2 \setminus \{(0,0)\}$ (Theorem 1.9),

it follows that, by Theorem 2.22, f is continuous on $\mathbb{R}^2 \setminus \{(0,0)\}$.

Final remark

1. $p(x,y) = (xy)^{1/3}$ is continuous at (0,0); and both p_x , p_y exist. We cannot apply Theorem 2.19, Corollary 2.20 or Theorem 2.22. We use Definition 2.14 to show that it is not differentiable at (0,0), i.e., $\lim_{(x,y)\to(0,0)} \frac{p(x,y)-L_{(0,0)}(x,y)}{\sqrt{(x-a)^2+(y-b)^2}}$ does not exist.
2. $f(x,y) = \frac{x^2y}{x^4+2y^2}$ is not continuous at (0,0). So by Corollary 2.20, it is not differentiable at (0,0). However, the question did not ask to show this.
3. $f(x,y) = -\frac{3xy}{x^2+y^2}$ is also not continuous at (0,0). So by Corollary 2.20, it is not differentiable at (0,0).